Description

Illustration



Figure caption

1. Decorative wood panelling (oak). 2. Hardwood used in structural and aesthetic applications. 3. Bridge made of solid oak. Images used under license from Shutterstock.com

The material

Hardwoods come from broad leave, deciduous, trees such as oak, ash, elm, sycamore, mahogany. Although most hardwoods are harder than softwoods, there are exceptions: balsa, for instance, is a hardwood. Wood must be seasoned before it is used. Seasoning is the process of drying the natural moisture out of the raw timber to make it dimensionally stable, allowing its use without shrinking or warping. In air-seasoning the wood is dried naturally in covered but open-sided structure. In kiln-drying the wood is artificially dried in an oven or kiln. Modern kilns are so designed that an accurate control of moisture is achieved.

Wood has been used for construction and to make products since the earliest recorded time. The ancient Egyptians used it for furniture, sculpture and coffins before 2500 BC. The Greeks at the peak of their empire (700 BC) and the Romans at the peak of theirs (around 0 AD) made elaborate buildings, bridges, boats, chariots and weapons of wood, and established the craft of furniture making that is still with us today. More diversity of use appeared in Mediaeval times, with the use of wood for large-scale building, and mechanisms such as pumps, windmills, even clocks, so that, right up to end of the 17th century, wood was the principal material of engineering. Since then cast iron, steel and concrete have displaced it in some of its uses, but timber continues to be used on a massive scale, particularly in housing and small commercial buildings. In 1997, 3.4 billion cubic meters of wood was consumed of which 1.9 billion cubic meters was burned as fuel (more than half). The combined value of this consumption reached 400 billion USD in the 1990s, about 2 percent of global GDP. Yet even with over half being consumed for fuel wood, more wood (by weight) is consumed in construction than either steel or concrete (LeVan 1998)

Compositional summary

Cellulose/Hemicellulose/Lignin/12%H2O

General properties

Density	850	-	1.03e3	kg/m^3

Price	* 5.06	-	8.15	GBP/kg
Date first used	-10000			

Material form that data applies to

Bulk	✓
Sheet	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	* 5	-	5.58	GPa
Shear modulus	* 0.517	-	0.71	GPa
Bulk modulus	* 2.52	-	2.82	GPa
Bending modulus	13.2	-	13.9	GPa
Poisson's ratio	* 0.02	-	0.04	
Yield strength (elastic limit)	* 4.26	-	5.22	MPa
Tensile strength	* 7.1	-	8.7	MPa
Compressive strength	* 12.8	-	15.6	MPa
Bending strength	70	-	95	MPa
Elongation	* 0.42	-	0.51	% strain
Hardness - Vickers	10	-	12.2	HV
Fatigue strength at 10^7 cycles	* 2.13	-	2.61	MPa
Fracture toughness	* 0.821	-	1	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.011	-	0.014	

Thermal and combustion properties

Thermal conductor or insulator?	Good insulator			
Thermal resistivity	5	-	6.13	m.°C/W
Thermal expansion coefficient	* 37.8	-	49.5	µstrain/°C
Specific heat capacity	1.66e3	-	1.71e3	J/kg.°C
Maximum service temperature	120	-	140	°C
Minimum service temperature	-73.2	-	-23.2	°C
Flammability	Highly flammable			
Emissivity	0.81	-	0.9	

Hygro-thermal properties

Water absorption	12	-	30	%
Frost resistance	Good			

Electrical properties

Electrical conductor or insulator?	Poor insulator				
Electrical resistivity	* 2.1e14	-	7e14	µohm.cm	
Dielectric constant (relative permittivity)	* 5.03	-	6.14		
Dissipation factor (dielectric loss tangent)	* 0.074	-	0.091		
Dielectric strength (dielectric breakdown)	* 1	-	2	MV/m	

Optical properties

Natural color

Brown

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Average
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Moldability	2	-	3
Machinability	5		

Durability

Water (fresh)	Limited use
Water (salt)	Limited use
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Good
Wear resistance	Unacceptable
Industrial atmosphere	Limited use

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Rural atmosphere	Acceptable			
Marine atmosphere	Limited use			

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.358	-	0.395	kg/kg
Embodied energy (virgin grade production)	* 68.1	-	75.1	MJ/kg
Water Usage	* 665	-	735	l/kg

Material processing: energy

Machining, coarse, embodied energy	* 15.5	-	17.1	MJ/kg
Machining, fine, embodied energy	* 17.2	-	19	MJ/kg
Machining, grinding, embodied energy	* 19	-	21	MJ/kg

Material processing: climate change

Machining, coarse, climate change (CO2-eq)	* 1.01	-	1.12	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.13	-	1.25	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.24	-	1.37	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.0215	-	0.0237	kg/kg
Embodied energy, recycling	* 0.681	-	0.751	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 19.8	-	21.3	MJ/kg
Combustion CO2	* 1.69	-	1.78	kg/kg
Landfill	✓			
Biodegrade	✓			
A renewable resource?	✓			

Supporting information

Design guidelines

Wood offers a remarkable combination of properties. It is light, and, parallel to the grain, it is stiff, strong and tough - as good, per unit weight, as any man-made material. It is cheap, it is renewable, and the fossil-fuel energy needed to cultivate and harvest it is outweighed by the energy it captures from the sun during growth. It is easily machined, carved and joined, and - when laminated - it can be molded to complex shapes. And it is aesthetically pleasing, warm both in color and feel, and with associations of craftsmanship and quality.

Technical notes

The values for the mechanical properties given for woods require explanation. Wood-science laboratories measure the mean properties of high-quality "clear" wood samples: small specimens with no knots or other defects; the data for woods in the database is of this type. This is not, however, the data needed for design. All engineering materials have some variability in quality and properties. To allow for this design handbooks list "allowables" - property values that will be met or exceeded by, say, 99% of all samples (meaning the mean value minus 2.33 standard deviations). Natural materials like wood show greater variability than man-made materials like steel, with the result that the allowable values for mechanical properties may be only 50% of the mean. There is a second problem: structures made of wood are much larger than the wood-science test samples. They contain knots, shakes and sloping grain, all of which degrade properties. To deal with this the wood is "stress-graded" by visual inspection or by automated methods, assigning each piece a stress grading G between 0 and 100: a grading of G means that properties are further knocked down by the factor G/100. Finally, in building construction, there is the usual requirement of sound practice - an overall safety factor, typically 2.25. The result is that the permitted stress for design may be as low as 20% of the value quoted in wood-science tabulations.
The data in this record is for oak of medium density, and lists wood-science ranges for the properties of clear wood samples.

Typical uses

Flooring; stairways, furniture; handles; veneer; sculpture, wooden ware; sash; doors; general millwork; framing- but these are just a few. Almost every load-bearing and decorative object has, at one time or another, been made from wood.

Related standards

BS 565 (1963)
BS 881 (1955)
BS 569 (1955)
BS 1860: Part 1 (1956)
CP 112 (1967)
CSI/CSC Masterformat Section Numbers:
06 10 00 ROUGH CARPENTRY
06 14 00 Treated Wood Foundations
06 16 00 Sheathing
06 12 00 Structural Panels
06 15 00 Wood Decking
06 17 00 Shop-Fabricated Structural Wood
06 17 53 Shop-Fabricated Wood Trusses
06 17 33 Wood I-Joists

Further reading

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American Wood Preservers Institute: www.preservedwood.com
Society of Wood Science and Technology: www.swst.org
Forest Products Society: www.forestprod.org

Links

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